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0.1-1 | Summary and Loads

This rivt file example calculates the maximum stress and deflection in a simply supported, uniformly loaded beam using E-B theory [0.1.1] . It also serves as an annotated example of a single rivt doc with multiple sections that is not part of a report.

The example illustrates the use of some of the most common API functions, commands and tags. Further details are provided in the rivt user manual .

The file may be formatted as a text, PDF or HTML doc by changing the type parameter in the Doc API | PUBLISH | command at the end of each rivt file (rv.D()). Published files are found in the _published folder.

0.1-2 | Load Combinations

Dead and live loads effects are taken from ASCE 7-05 [0.1.2]

Table 2.1: Load Effects (stored: t001-1.csv)

Equation No.	Load Combination
16-1	1.4(D+F)
16-2	1.2(D+F+T) + 1.6(L+H) + 0.5(Lr or S or R)
16-3	1.2(D+F+T) + 1.6(Lr or S or R) + (f1L or 0.8W)

0.1-3 | Loads and Geometry

Successive value definitions are formatted as a table. Variable values are defined with the define operator. The line tag [T] labels and numbers the table.

Table 3.1: Define Unit Loads

variable	value	[value]	description
D_1	3.80 p_sf	0.18 kPA	joists DL
D_2	2.10 p_sf	0.10 kPA	plywood DL
D_3	10.00 p_sf	0.48 kPA	partitions DL
D_4	3.00 k_ft	43.78 kN_m	fixed machinery DL
L_1	40.00 p_sf	1.92 kPA	ASCE7-O5 LL

variable	value	[value]	description
b_1	10.00 inch	254.00 mm	beam width
h_1	18.00 inch	457.20 mm	beam depth
E_1	29000.00 k_si	199947.96 MPA	modulus of elasticity
Fb_1	20000.00 p_si	137.90 MPA	allowable stress

The VALTABLE command reads variable values from a file in the rvsrc/data folder. The description is the table title, followed by the max column width.

Table 3.2: Beam Geometry (beam1.csv)

variable	value	[value]	description
spc_1	2.00 ft	0.61 m	beam spacing
spn_1	16.00 ft	4.88 m	beam span

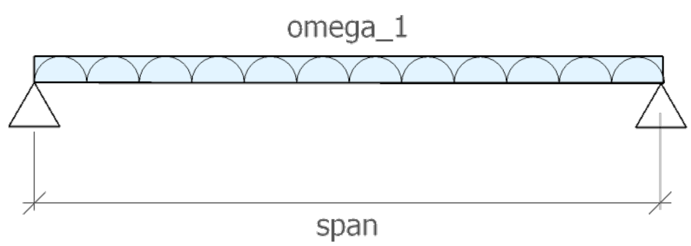

Fig. 3.1 - Beam Diagram

Uniform Distributed Loads

Eq. 3.1: Dead load [ASCE7-05 2.3.2]

$$dl_1 = 1.2 \cdot (D_4 + spc_1 \cdot (D_1 + D_2 + D_3))$$

dl_1 = 3.64 k_ft   [dl_1] = 53.09 kN_m | Dead load [ASCE7-05 2.3.2]

D_4	spc_1	D_2	D_3	D_1
3.00 k_ft	2.00 ft	2.10 p_sf	10.00 p_sf	3.80 p_sf
fixed machinery DL	beam spacing	plywood DL	partitions DL	joists DL

Eq. 3.2: Live load [ASCE7-05 2.3.2]

$$ll_1 = 1.6 \cdot L_1 \cdot spc_1$$

ll_1 = 0.13 k_ft [ll_1] = 1.87 kN_m | Live load [ASCE7-05 2.3.2]

L_1	spc_1
40.00 p_sf	2.00 ft
ASCE7-05 LL	beam spacing

Eq. 3.3: Total load [ASCE7-05 2.3.2]

$$\omega_1 = dl_1 + ll_1$$

ω_1 = 3.77 k_ft [ω_1] = 54.96 kN_m | Total load [ASCE7-05 2.3.2]

dl_1	ll_1
3.64 k_ft	128.00 ft·p_sf
Dead load [ASCE7-05 2.3.2]	Live load [ASCE7-05 2.3.2]

0.1-4 | Beam Response

The following lines import the beam geometry from an external file, calculate section properties from imported functions and calculate the maximum moment, bending stress and mid-span deflection.

Table 4.1: Beam functions (sectprop.py)

Function	Docstring
rectsect(b, d)	section modulus of rectangle
rectinertia(b, d)	moment of inertia of rectangle
midspan_delta(ln, w, e, i)	mid-span deflection of simply supported beam with UDL

Eq. 4.2: rectangle - S (sectprop.py)

$section_1 = rectsect(b_1, h_1)$

$section_1$ = 540.00 in3 [$section_1$] = 8849.01 cm3 | rectangle - S (sectprop.py)

b_1	h_1
10.00 inch	18.00 inch
beam width	beam depth

Eq. 4.3: rectangle - I (sectprop.py)

$inertia_1 = rectinertia(b_1, h_1)$

$inertia_1$ = 4860.0 in4 [$inertia_1$] = 202288.5 cm4 | rectangle - I (sectprop.py)

b_1	h_1
10.0 inch	18.0 inch
beam width	beam depth

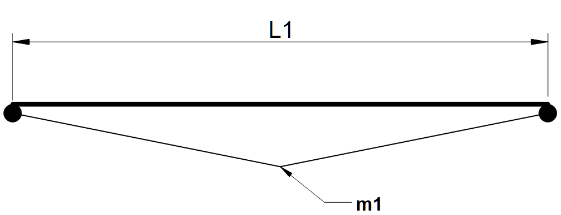

Fig. 4.1 - Moment diagram

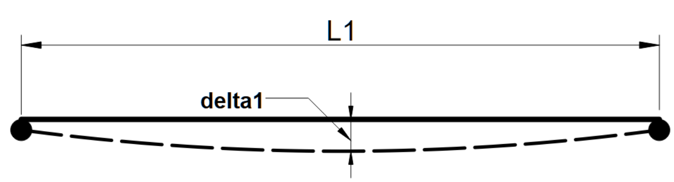

Fig. 4.2 - Deflection diagram

Maximum bending stress formula

Eq.4.4:

$$\sigma_1 = \frac{M_1}{S_1}$$

Eq. 4.4: Mid-span UDL moment

$$m_1 = \frac{\omega_1 \cdot spn_1^2}{8}$$

```
m₁ = 120.52 ft-kip     [m₁] = 163.40 mkN    | Mid-span UDL moment
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spn_1	ω_1
16.00 ft	3.77 k_ft
beam span -	Total load [ASCE7-05 2.3.2]

Eq. 4.5: Bending stress

$$fb_1 = \frac{m_1}{section_1}$$

```
fb₁ = 2678.2 p_si     [fb₁] = 18.5 MPA    | Bending stress
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$section_1$	m_1
540.0 inch3	120.5 ft2·k_ft
rectangle - S (sectprop.py)	Mid-span UDL moment -

Eq.4.7: Stress ratio

$fb_1 < Fb_1$

[1] fb_1	[2] Fb_1	ratio [1]/[2]	check	reference
2.68 k_si	20.00 k_si	0.13	OK	Stress ratio

Eq. 4.8: mid-span deflection (sectprop.py)

```
δ₁ = midspan_δ(spn₁, ω₁, E₁, inertia₁)

δ₁ = 0.04 inch     [δ₁] = 1.00 mm   | mid-span deflection (sectprop.py)
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spn₁	ω₁	inertia₁	E₁
16.00 ft	3.77 k_ft	4860.00 inch4	29000.00 k_si
beam span -	Total load [ASCE7-05 2.3.2]	rectangle - I (sectprop.py)	modulus of elasticity

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[0.1.1]

"Euler–Bernoulli beam theory", Wikipedia, Wikimedia Foundation. [Online].https://en.wikipedia.org/wiki/Euler_Bernoulli_beam_theory.[Accessed: Jun. 15, 2026].

[0.1.2]

ASCE/SEI 7-05, Minimum Design Loads for Buildings and Other Structures,American Society of Civil Engineers, 2005.